HACKURITY 2026 · PS-6 · TRACK C

Chain of Custody

Tamper-evident logging that stops, detects and contains attacks on the evidence itself.

Layer 1 · Stop | Layer 2 · Detect | Layer 3 · Contain

Firewall Crew
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WHY THIS MATTERS

Logs are the evidence

If the logs can be changed, nothing they prove can be trusted.

Banking & fintech

Every transaction, approval and balance change. Audit trails regulators check.

SOC & incident response

Logs are how an attack is found, scoped and replayed.

Critical infrastructure

Power, telecom, water. Every control command must be traceable.

Healthcare

Who opened which patient record, and when.

SaaS & cloud

Admin actions, access changes, customer data reads.

Legal & compliance

Court evidence needs an unbroken chain. CERT-In: 180-day log retention.

THE ATTACKER'S VIEW

Why attackers go after logs

Wiping the trail is a standard step after a break-in. MITRE ATT&CK lists it as **T1070, Indicator Removal**.

01 **Hide the break-in**
Erase how they got in and what they ran.

02 **Stay inside longer**
No trail, no alert, more time on the system.

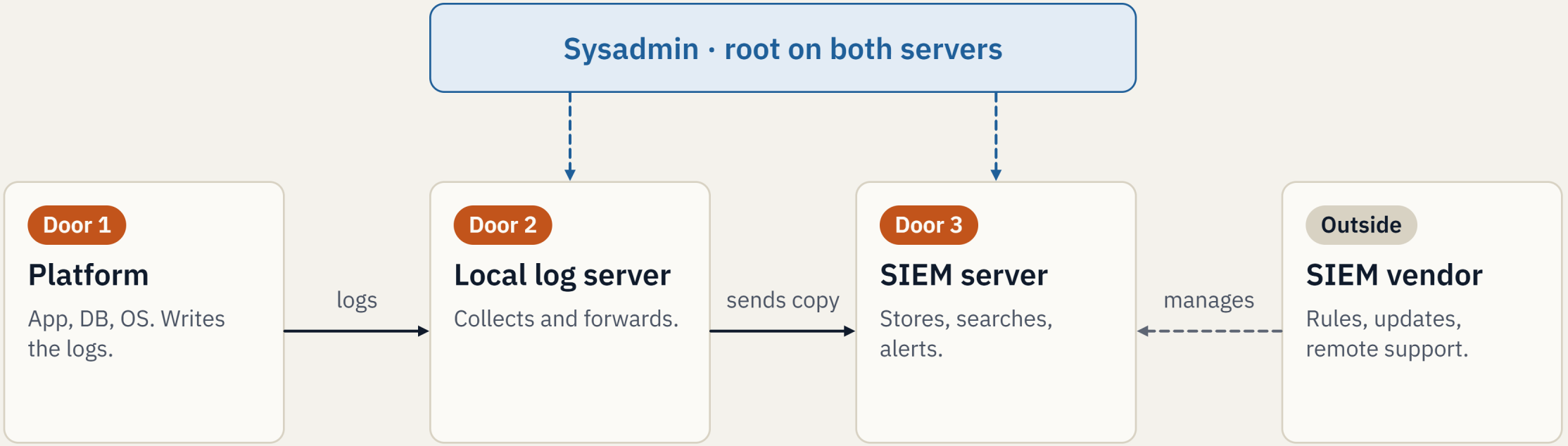
03 **Commit insider fraud**
Change an amount, then change the log to match.

04 **Break the evidence**
One gap or edit makes the whole log useless in court.

05 **Blind the SOC**
Kill the forwarder. The SIEM reads silence as a quiet night.

How it works today

Every hop has an admin. Every admin root is a door.



Attacker needs one admin root. Then they edit at the source, in transit, or inside the SIEM.

What else breaks today

Beyond the three doors, the pipeline itself has blind spots.

Edit before ship

Changed in the gap between write and forward.

Silent agent kill

Forwarder stops. Silence looks normal.

SIEM trusts input

No proof an entry is original. It stores what arrives.

Box-supplied time

Timestamps come from the machine under attack.

Truncate & rotate

Cut the tail, or abuse log rotation to drop entries.

Shared admin roots

Same people, same keys across platform and SIEM.

Log injection

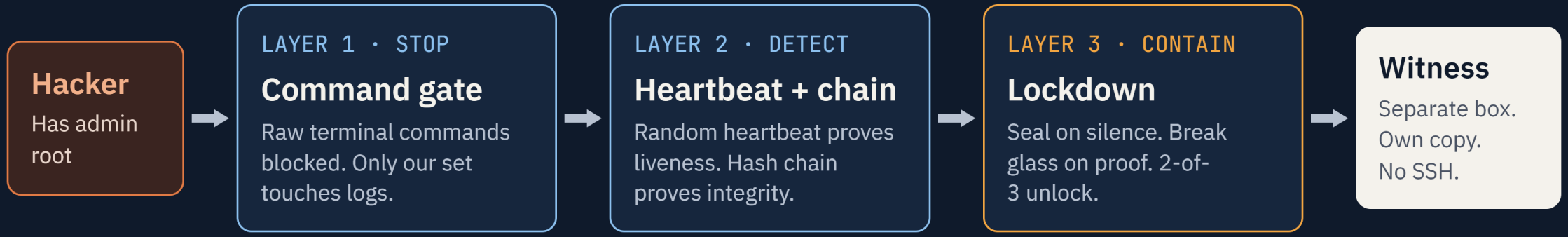
Fake lines slipped in through user input fields.

Delete & replay

Remove entries, replay old ones to fill the gap.

OUR ARCHITECTURE

Three layers, one witness

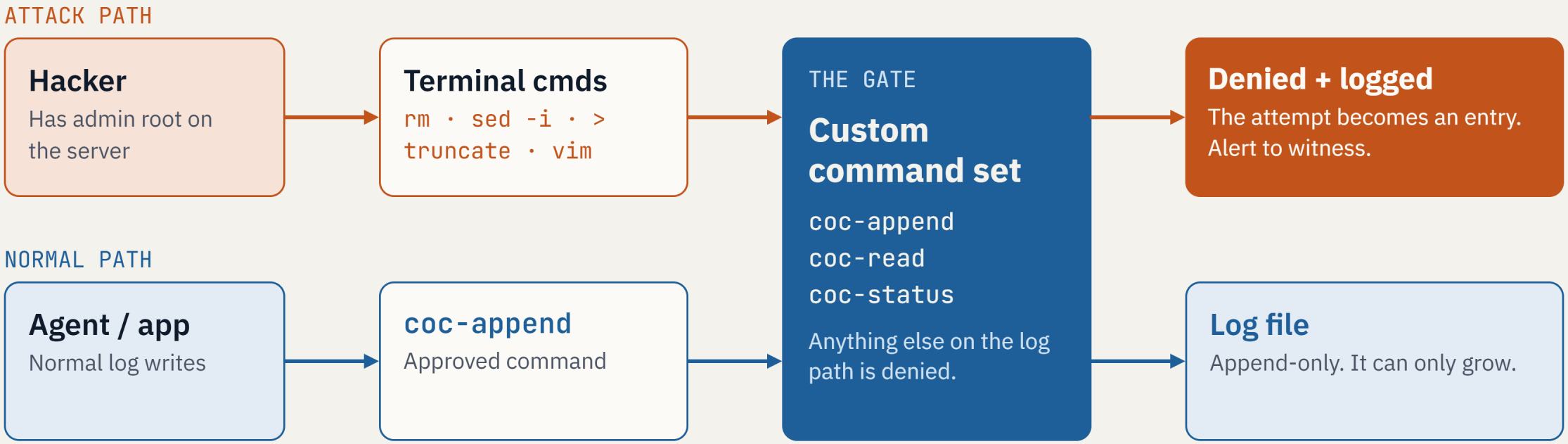


Assume each layer can fall. The next one still catches it.

Alerts only appear on the witness, never on the box the hacker controls.

LAYER 1 · STOP

Command gate on the log server

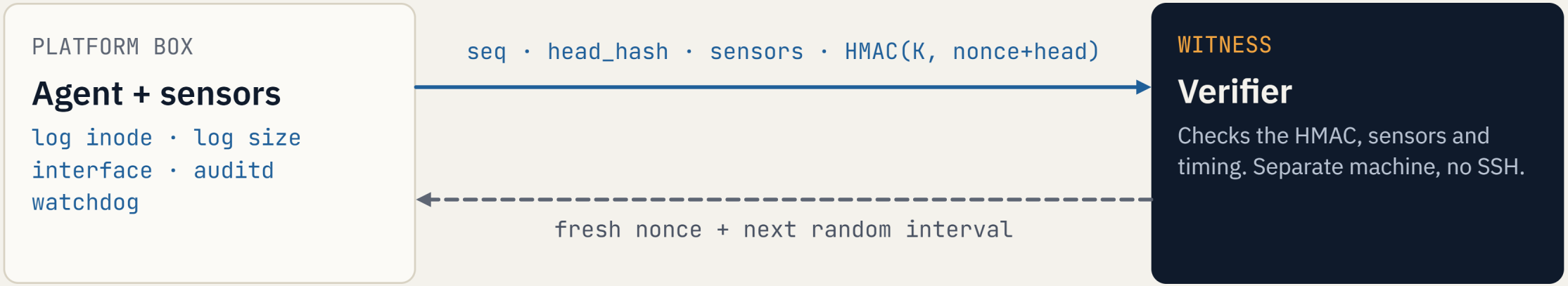


append-only attribute (chattr +a) · command allowlist wrapper · no raw write path · every attempt is evidence

LAYER 2A · DETECT

Dynamic heartbeat

Assume layer 1 fell. The box must keep proving it’s alive and untouched.



Random interval

Jittered timing. No predictable window to tamper in.

Fresh nonce

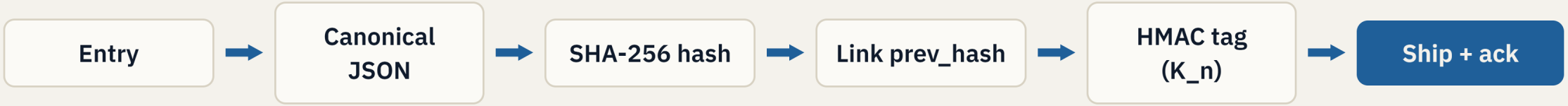
New challenge every beat. Can’t replay or precompute.

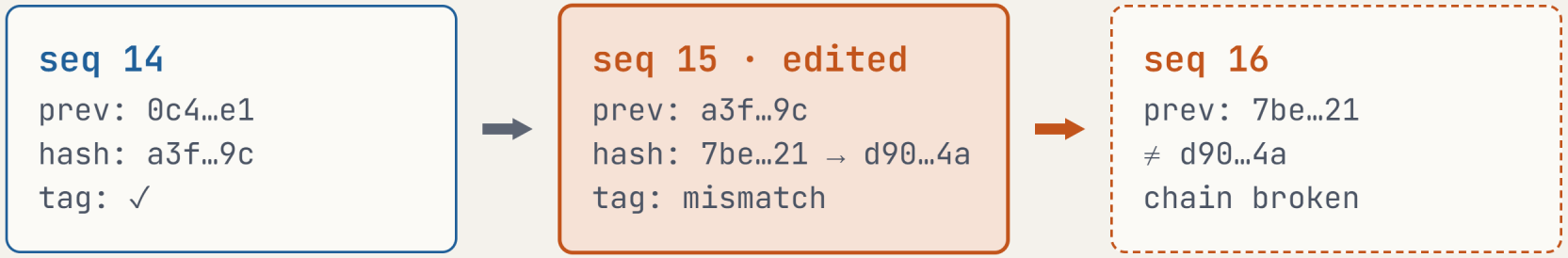
3 missed beats

Silence alert with the last acked seq.

LAYER 2B · DETECT

Checksum chain





Change one entry, the chain breaks at that exact seq. Re-signing fails: old keys are gone.

key ratchet every 100 entries

old key deleted (forward-secure)

balance reconciliation checksum

only acked entries count as safe

LAYER 3 · CONTAIN

Lockdown, matched to the evidence

TRIGGER

Signal lost

Network cut or missed beats

→

RESPONSE

Seal

Log goes read-only. Agent keeps buffering.

→

RECOVERY

Restore

Flush from acked_seq + 1. Exact gap reported.

TRIGGER

Proof of tampering

Tag mismatch, chain break, L1 bypass

→

RESPONSE

Break glass

Freeze. Lock out hacker and admin. Evidence copy to witness.

→

RECOVERY

2-of-3 unlock

Split key. No single person can reopen it.

witness: no SSH | ingest-only API | Shamir-style 2-of-3 shares | a cut seals, it never wipes

COVERAGE MAP

Every attack, the layer that catches it

Attack	Caught by	How
rm / sed on the log	Layer 1	Gate denies it, attempt logged
Naive payload edit	Layer 2b	Hash mismatch at the exact seq
Rewrite with stolen key	Layer 2b	Tag mismatch: old keys are deleted
Truncate last lines	Layer 2a + 2b	Size sensor drops, witness copy still has them
Kill the agent	Layer 2a	Silence alert with last acked seq
Network cut	Layer 2a + 3	Silence alert, seal, exact gap count on restore
Interface toggled	Layer 2a	Box sensor alert, no false silence alarm
Confirmed tampering	Layer 3	Break glass, 2-of-3 unlock
Edit before first ship	Known limit	Witness receives the original outbox copy

TRACK C · BLOCKCHAIN

The building blocks we borrowed

Blockchain’s integrity ideas, without a chain-wide network to run.

Hash-linked blocks	→	**Every entry carries prev_hash**
Signed transactions	→	**HMAC tag per entry, forward-secure keys**
Independent nodes hold copies	→	**Witness keeps its own copy, no SSH**
Confirmation / finality	→	**Witness ack: only acked entries are safe**
Immutability	→	**Append-only + detect any change at its seq**

CASE STUDY

Why big enterprise ledgers were scrapped

AUSTRALIA · STOCK SETTLEMENT

ASX CHESS replacement

Goal: replace the 1990s clearing and settlement system with a DLT platform.

Ended: scrapped Nov 2022 after years of delays. About A$250M written off.

Why: scope kept growing; an independent review found the design wouldn't meet requirements.

GLOBAL SHIPPING · MAERSK + IBM

TradeLens

Goal: one shared ledger for shipping documents and container events.

Ended: launched 2018, shut down early 2023.

Why: rival carriers wouldn't join a Maersk-led platform. No network, no viability.

Lesson: the hash chain was never the hard part. Scope, adoption and trust in the inputs were.

THE GAPS WE COVER

Why it’s still rare, and our answer

Why blockchain logging stalls	What Chain of Custody does
Oracle problem: proves nothing about the input	Guards the source: L1 gate, sensors, heartbeat
Heavy infra and consensus cost	Hash chain + one witness, plain Python
Needs the whole industry on board (TradeLens)	Works for one org, per-platform profiles
Rip-and-replace legacy (ASX)	Bolts onto existing logs; the SIEM stays
Immutable, but no clean way to correct	Detect, don’t forbid: a fix is a new entry
Detects, but has no response	Layer 3: seal or break glass

LIVE DEMO

What you'll see

1
Normal run
All verified, zero alerts

2
Hacker tries rm
Layer 1 denies it

3
Edit seq 15
Layer 2 flags the exact seq

4
Cut the network
Silence, seal, exact gap on restore

5
Break glass
Lockout, then 2-of-3 unlock

Logs that can prove nobody touched them.

Watch the witness view: it shows which layer caught each attack.